Wall time per step: Parse 1M PBMC, 100k (69,412 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	10	6.2	5.6	5.4	19	2.3	6.1	38	6.5	72	43	10	15
qc	0.6	0.8	0.4	0.4	0.7	0.8	1.1	0.5	0.5	1.1	2.6	2.7	1.2
normalise	n/a	n/a	n/a	n/a	2.0	1.9	0.0	0.1	0.1	0.0	0.1	0.1	0.5
hvg	0.3	0.3	0.3	0.2	1.4	1.4	1.7	0.2	0.2	2.7	4.0	4.0	4.0
pca	0.7	0.6	0.5	0.3	1.4	1.4	4.0	1.7	1.7	4.1	7.1	6.3	73
harmony	0.6	0.5	0.5	0.4	0.6	0.6	0.5	12	12	13	13	13	14
knn_snn	2.0	2.1	1.1	1.1	27	27	24	4.3	4.6	4.4	5.3	5.4	5.5
cluster	0.8	0.8	0.7	0.7	2.0	2.0	1.9	8.8	8.7	8.4	18	18	18
cluster_fast	0.7	0.8	0.4	0.2	n/a	n/a	n/a	10	11	10	n/a	n/a	n/a
umap	2.6	2.5	0.5	0.5	27	27	26	32	33	33	35	35	35
dge	0.7	0.7	0.7	0.6	56	52	–	6.2	6.3	63	15	16	196
tsne_default	27	26	26	24	96	95	102	n/a	n/a	n/a	66	71	71
tsne_fft	8.4	7.7	2.3	2.1	32	32	31	n/a	n/a	n/a	24	24	24
tsne_quick	5.1	4.8	3.1	2.7	n/a	n/a	n/a	17	18	17	n/a	n/a	n/a
total (no tSNE)	19	15	10	9.7	137	116	65	104	73	201	143	110	361

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.